Python computing lab course

Engineering Communication, Measurement, and Data Analysis

Session 4: NumPy, MatPlotLib and Pandas

Objectives

After completing these exercises you will be able to:

- Import and use number python (NumPy)
- Use pandas to import and output data
- Use Matlibplot to plot data on line and scatter plots

Question 1 (core): Strain-stress graph

Please follow these steps to create a stress-strain graph and generate a new data file (`.csv`) containing the calculated stress and strain values.

1. Download the file `ForceDisplacementAl.csv` from Blackboard. This file contains force and displacement data from a tensile test on an aluminium sample.
2. **Open the `.csv` file using pandas.**
3. Extract the data into NumPy arrays and use the following variable names:
   - `Force`
   - `Displacement`
4. The material has the following dimensions. Use these values to calculate the cross-sectional area of the sample:
   - `Diameter = 10 mm`
   - `Original Length = 50 mm`

   To calculate the cross-sectional area, use this formula:

$$\text{Cross-sectional Area} = \pi \left(\frac{\text{Diameter}}{2} \right)^2$$

5. Now calculate the stress (MPa) and strain (%) using the formulas below:

$$\text{Stress} = \frac{\text{Force}}{\text{Cross-sectional Area}}$$

$$\text{Strain} = \frac{\text{Displacement}}{\text{Original Length}} \times 100\%$$

6. Create a graph that shows stress on the y-axis and strain on the x-axis. Make sure the axes are labelled correctly.
7. Save the graph as an image file with a DPI (dots per inch) setting of 400.
8. Find and print the maximum stress value. This is the tensile strength of the material.
9. Find and print the maximum strain value. This shows the strain at the point of failure.
10. Finally, save the stress and strain data to a new `.csv` file.

Question 2 (core): Your thermodynamics experiment

In this exercise you will analyse your results from thermodynamics experiment. Table 1 shows an example of how you might have presented your data in your lab book.

Table 1: Temperature and displacement under various environmental conditions

Environment	Temperature (°K)	Syringe offset (mm)
Room Temperature		
Hot 1		
Hot 2		
Hot 3		
Cold 1		
Cold 2		
Cold 3		

The specific gas constant can be estimated from these measurements using a rearranged form of the ideal gas law:

$$V = \frac{mR}{p}T \tag{1}$$

and comparing it to the equation for a straight line:

$$y = ax + b \tag{2}$$

As long as the same mass of air is used for all of your V, T measurements, they will lie on the same straight line. As density and pressure are known the gradient of this straight line can be used to estimate R.

You will import, process and plot your data collected from the thermodynamics experiment. Below includes the steps you should follow to create a plot to be used in your report.

If you have not collected data yet you can follow these steps using ThermoExampleData_Argon.csv for now (note that this data is for argon, not air), and apply your code to your data when you have gathered it.

1. Create a .csv file of your data, using the layout shown in Table 1 to help you structure your data in excel. A .csv file can be created by inputting your data into an excel spreadsheet and then saving a copy of the file using save as .csv file
2. Using pandas import the data and extract temperature and offset numpy arrays
3. Create a numpy array of volumes from your offsets (using known syringe radius)
4. Calculate the gradient of V against T using the following formula for the linear regression of a straight line with intercept equal to zero.

$$a = \frac{\sum x_i y_i}{\sum (x_i^2)} \tag{3}$$

5. Plot your V, T data and add a line of best fit with gradient a and intercept b
6. Add axis labels (with units) and a legend to your plot
7. Save a .png of your plot to use in your report
8. Calculate the mass of air in the syringe using an appropriate room temperature density value
9. Calculate and print your value of R (with units and 3 sig. figs.)

Question 3 (extension): Evaluating Hertz contact stresses

Hertz contact stresses occur when two spheres (of elastic material) are pressed together with force F as shown in figure 1. The following series of expressions describe the principal contact stresses in

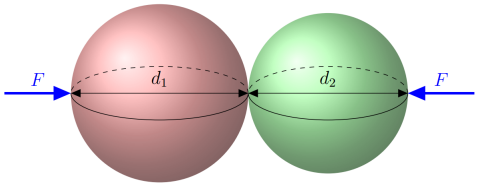


Figure 1: Contacts of spheres

the x-, y- and z-directions, respectively, at the contact point of the spheres.

$$\sigma_x = \sigma_y = -p_{max}\left[\left(1 - \frac{z}{a}\tan^{-1}\left(\frac{a}{z}\right)\right)(1-\nu_1) - \frac{1}{2}\left(1+\frac{z^2}{a^2}\right)^{-1}\right] \tag{4}$$

$$\sigma_z = \frac{-p_{max}}{1+z^2/a^2} \tag{5}$$

Where a is the radius of the contact point after elastic deformation and p_{max} is the maximum contact pressure, which are given by:

$$a = \sqrt[3]{\frac{3F}{8}\frac{(1-v_1^2)/E_1 + (1-v_2^2)/E_2}{1/d_1 + 1/d_2}} \tag{6}$$

$$p_{max} = \frac{3F}{2\pi a^2} \tag{7}$$

Where ν_1, E_1 and d_1 are the Poisson's ratio, Young's modulus and diameter of the first sphere and ν_2, E_2 and d_2 are for the second sphere.
Note: the cubic root of a variable can be found by using **np.cbrt()**.

Write a program which determines principal stresses in the x,y and z direction for values of z between 0.20 and 0.30mm with intervals of 0.01 between values. The values of $\nu_1 = \nu_2 = 0.3$, $E_1 = E_2 = 206GPa$, $d_1 = 38mm$, $d_2 = 70mm$ and $F = 450N$.

The program should output values of stresses in x, y and z direction (as a NumPy arrays) and plot a line graph of z vs. stress in x. The expected results are shown in figure 2, your plot should be formatted correctly like the one shown in the figure.

Expected values:-
$(1-v_1^2)/E = 4.4175 \times 10^{-6}$ (same value for $(1-v_2^2)/E$)
$p_{max} = 1944.8376$

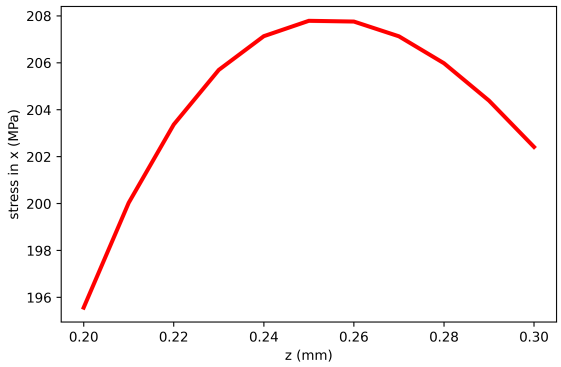


Figure 2: Line plot of the contact stress in the x direction at various z positions

Question 4 (extension): Plotting of temperature across a square plate

A square plate is heated at a single point on the underside of the plate, the temperature at random points on the top surface are measured and their xy coordinates and temperature in Fahrenheit is record in the file *Heat map.csv*.

$$T_{\text{Kelvin}} = (T_{\text{Fahrenheit}} - 32) \times 5/9 + 273.15 \tag{8}$$

Equation 8 shows the conversion of temperature in Fahrenheit to Kelvin.

Task 2
Write a script that will import data from *Heat map.csv*, converts the temperature from Fahrenheit to Kelvin, plots a scatter plot, saves the image and then output the data to a new file called *Heat map in K.csv*. The scatter plot should have an appropriate title, labelled x and y axis (with units of mm) and a labelled colour bar. The image should be saved as a The headers for the output file should be x coordinate (mm); y coordinate (mm); and Temperature (K).

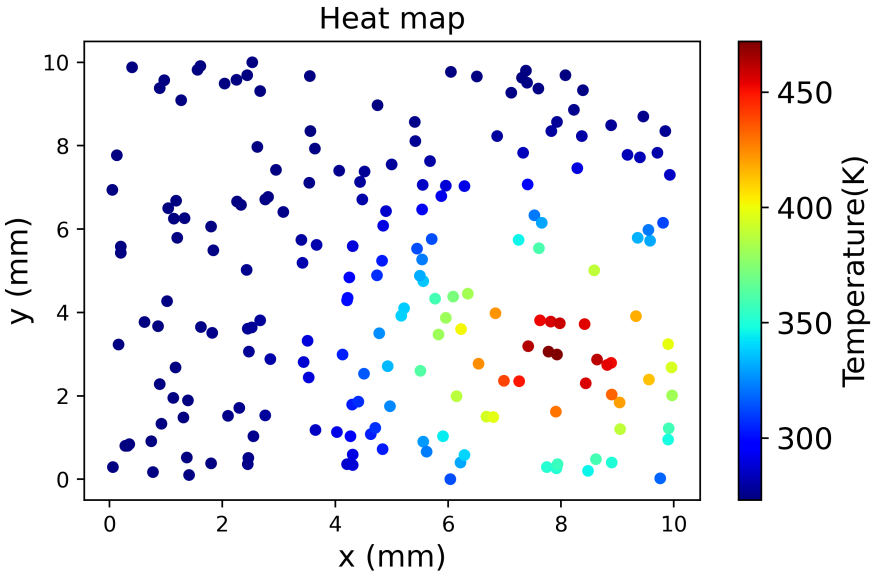


Figure 3: Example plot: Scatter plot of temperature across a plate